Q1 - 2024 (01 Feb Shift 1)

Let S $= \left\{x \in R : (\sqrt{3}+\sqrt{2})^x + (\sqrt{3}-\sqrt{2})^x = 10\right\}$.

Then the number of elements in S is :

(1) 4

(2) 0

(3) 2

(4) 1

Q2 - 2024 (01 Feb Shift 2)

Let α and β be the roots of the equation $\mathrm{px}^2 + \mathrm{qx} - r = 0$, where $p \neq 0$. If p, q and r be the consecutive terms of a non-constant G.P and $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3}{4}$, then the value of $(\alpha - \beta)^2$ is :

(1) $\frac{80}{9}$

(2) 9

(3) $\frac{20}{3}$

(4) 8

Q3 - 2024 (27 Jan Shift 2)

If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then

(1) $2\mathrm{~S}_{12} = \mathrm{S}_{11} + \mathrm{S}_{10}$

(2) $\mathrm{S}_{12} = \mathrm{S}_{11} + \mathrm{S}_{10}$

(3) $2\mathrm{~S}_{11} = \mathrm{S}_{12} + \mathrm{S}_{10}$

(4) $\mathrm{S}_{11} = \mathrm{S}_{10} + \mathrm{S}_{12}$

Q4 - 2024 (29 Jan Shift 2)

Let the set $C = \left\{(x, y) \mid x^2 - 2^y = 2023, x, y \in \mathbb{N}\right\}$. Then $\sum_{(x,y)=C}(x+y)$ is equal to ______

Q5 - 2024 (30 Jan Shift 1)

Let $\alpha, \beta \in \mathrm{N}$ be roots of equation $\mathrm{x}^2 - 70\mathrm{x} + \lambda = 0$, where $\frac{\lambda}{2}, \frac{\lambda}{3} \notin \mathrm{N}$. If λ assumes the minimum possible value, then $\frac{(\sqrt{\alpha-1}+\sqrt{\beta-1})(\lambda+35)}{|\alpha-\beta|}$ is equal to:

Q6 - 2024 (30 Jan Shift 2)

The number of real solutions of the equation $x\left(x^2+3|x|+5|x-1|+6|x-2|\right)=0$ is________.

Q7 - 2024 (31 Jan Shift 1)

For $0 < \mathrm{c} < \mathrm{b} < \mathrm{a}$, let $(\mathrm{a}+\mathrm{b}-2\mathrm{c})\mathrm{x}^2 + (\mathrm{b}+\mathrm{c}-2\mathrm{a})\mathrm{x} + (c+a-2b) = 0$ and $\alpha \neq 1$ be one of its root.

Then, among the two statements

(I) If $\alpha \in (-1, 0)$, then b cannot be the geometric mean of a and c

(II) If $\alpha \in (0, 1)$, then b may be the geometric mean of a and c

(1) Both (I) and (II) are true

(2) Neither (I) nor (II) is true

(3) Only (II) is true

(4) Only (I) is true

Q8 - 2024 (31 Jan Shift 1)

Let S be the set of positive integral values of a for which $\frac{\mathrm{ax}^2+2(\mathrm{a}+1)\mathrm{x}+9\mathrm{a}+4}{\mathrm{x}^2-8\mathrm{x}+32} < 0, \forall \mathrm{x} \in \mathbb{R}$.

Then, the number of elements in S is :

(1) 1

(2) 0

(3) ∞

(4) 3

Q9 - 2024 (31 Jan Shift 2)

The number of solutions, of the equation $\mathrm{e}^{\sin \mathrm{x}} - 2\mathrm{e}^{-\sin \mathrm{x}} = 2$ is

(1) 2

(2) more than 2

(3) 1

(4) 0

Q10 - 2024 (31 Jan Shift 2)

Let a, b, c be the length of three sides of a triangle satisfying the condition $\left(a^2+b^2\right) x^2-2b(a+c)$. $x+\left(b^2+c^2\right)=0$. If the set of all possible values of x is the interval (α, β), then $12\left(\alpha^2+\beta^2\right)$ is equal to ________.

Answer Key

Q1 (3)	**Q2** (1)	**Q3** (2)	**Q4** (46)
Q5 (60)	**Q6** (1)	**Q7** (1)	**Q8** (2)
Q9 (4)	**Q10** (36)		

Q1

$(\sqrt{3}+\sqrt{2})^x+(\sqrt{3}-\sqrt{2})^x=10$

Let $(\sqrt{3}+\sqrt{2})^{\mathrm{x}}=\mathrm{t}$

$\mathrm{t}+\frac{1}{\mathrm{t}}=10$

$\mathrm{t}^2-10\mathrm{t}+1=0$

$t=\frac{10\pm\sqrt{100-4}}{2}=5\pm2\sqrt{6}$

$(\sqrt{3}+\sqrt{2})^x=(\sqrt{3}\pm\sqrt{2})^2$

$\mathrm{x}=2$ or $\mathrm{x}=-2$

Number of solutions $=2$

Q2

$\mathrm{px}^2+\mathrm{qx}-\mathrm{r}=0<_{\beta}^{\alpha}$

$p=A, q=AR, r=AR^2$

$Ax^2+ARx-AR^2=0$

$x^2+Rx-R^2=0<_{\beta}^{\alpha}$

$$\because \frac{1}{\alpha}+\frac{1}{\beta}=\frac{3}{4}$$

$$\therefore \frac{\alpha+\beta}{\alpha\beta}=\frac{3}{4} \Rightarrow \frac{-\mathrm{R}}{-\mathrm{R}^2}=\frac{3}{4} \Rightarrow \mathrm{R}=\frac{4}{3}$$

$$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=\mathrm{R}^2-4\left(-\mathrm{R}^2\right)=5\left(\frac{16}{9}\right)$$

$=80/9$

Q3

$\mathrm{x}^2-\mathrm{x}-1=0$

$\mathrm{S}_{\mathrm{n}}=2023\alpha^{\mathrm{n}}+2024\beta^{\mathrm{n}}$

$\mathrm{S}_{\mathrm{n}-1}+\mathrm{S}_{\mathrm{n}-2}=2023\alpha^{\mathrm{n}-1}+2024\beta^{\mathrm{n}-1}+2023\alpha^{\mathrm{n}-2}+2024\beta^{\mathrm{n}-2}$

$=2023\alpha^{\mathrm{n}-2}[1+\alpha]+2024\beta^{\mathrm{n}-2}[1+\beta]$

$=2023\alpha^{\mathrm{n}-2}\left[\alpha^2\right]+2024\beta^{\mathrm{n}-2}\left[\beta^2\right]$

$=2023\alpha^{\mathrm{n}}+2024\beta^{\mathrm{n}}$

$\mathrm{S}_{\mathrm{n}-1}+\mathrm{S}_{\mathrm{n}-2}=\mathrm{S}_{\mathrm{n}}$

Put $\mathrm{n}=12$

$\mathrm{S}_{11}+\mathrm{S}_{10}=\mathrm{S}_{12}$

Q4

$x^2 - 2^y = 2023$
$\Rightarrow x = 45, y = 1$
$\sum_{(x,y)=C} (x+y) = 46.$

Q5

$x^2 - 70x + \lambda = 0$

$\alpha + \beta = 70$

$\alpha\beta = \lambda$

$\therefore \alpha(70 - \alpha) = \lambda$

Since, 2 and 3 does not divide λ

$\therefore \alpha = 5, \beta = 65, \lambda = 325$

By putting value of α, β, λ we get the required value 60 .

Q6

$x = 0$ and $x^2 + 3|x| + 5|x-1| + 6|x-2| = 0$

Here all terms are +ve except at $x = 0$

So there is no value of x

Satisfies this equation

Only solution x $= 0$

No of solution 1 .

Q7

$f(x)=(a+b-2c)x^2+(b+c-2a)x+(c+a-2b)$
$f(x)=a+b-2c+b+c-2a+c+a-2b=0$
$f(1)=0$
$\therefore \alpha \cdot 1=\dfrac{c+a-2b}{a+b-2c}$
$\alpha=\dfrac{c+a-2b}{a+b-2c}$
If $-1<\alpha<0$
$-1<\dfrac{c+a-2b}{a+b-2c}<0$
$b+c<2a$ and $b>\dfrac{a+c}{2}$
therefore, b cannot be G.M. between a and c.

If, $0<\alpha<1$

$0<\frac{c+a-2b}{a+b-2c}<1$

$b>c$ and $b<\frac{a+c}{2}$

Therefore, b may be the G.M. between a and c.

Q8

$ax^2+2(a+1)x+9a+4<0 \forall x \in R$
$\therefore a<0$

Q9

Take $e^{\sin x}=t(t>0)$
$\Rightarrow t-\dfrac{2}{t}=2$
$\Rightarrow \dfrac{t^2-2}{t}=2$
$\Rightarrow t^2-2t-2=0$
$\Rightarrow t^2-2t+1=3$
$\Rightarrow (t-1)^2=3$
$\Rightarrow t=1\pm\sqrt{3}$
$\Rightarrow t=1\pm1.73$
$\Rightarrow t=2.73$ or -0.73 (rejected as $t>0$)
$\Rightarrow e^{\sin x}=2.73$
$\Rightarrow \log_e e^{\sin x}=\log_e 2.73$
$\Rightarrow \sin x=\log_e 2.73>1$
So no solution.

Q10

$\left(a^2+b^2\right) x^2-2b(a+c)x+b^2+c^2=0$
$\Rightarrow a^2x^2-2abx+b^2+b^2x^2-2bcx+c^2=0$
$\Rightarrow (ax-b)^2+(bx-c)^2=0$
$\Rightarrow ax-b=0, \quad bx-c=0$
$\Rightarrow a+b>c \quad b+c>a \quad c+a>b$
$a+ax>bx \quad |ax+bx>a|ax^2+a>ax$

$a+ax>ax^2 \left| \begin{array}{l} ax+ax^2>a \\ x^2-x+1>0 \end{array} \right.$

$x^2-x-1<0 \left| x^2+x-1>0 \right|$ always true

$\dfrac{1-\sqrt{5}}{2}<x<\dfrac{1+\sqrt{5}}{2}$

$x<\dfrac{-1-\sqrt{5}}{2}$, or $x>\dfrac{-1+\sqrt{5}}{2}$

$\Rightarrow \dfrac{\sqrt{5}-1}{2}<\mathrm{x}<\dfrac{\sqrt{5}+1}{2}$

$\Rightarrow \alpha=\dfrac{\sqrt{5}-1}{2}, \beta=\dfrac{\sqrt{5}+1}{2}$

$2\left(\alpha^2+\beta^2\right)=12\left(\dfrac{(\sqrt{5}-1)^2+(\sqrt{5}+1)^2}{4}\right)=36$